1. Class

Program 1: Student details

```
#include <iostream>
using namespace std;
class Student {
public:
    string name; int rollNo; float marks;
    void display() {
        cout << "Name: " << name << "\nRoll No: " << rollNo
             << "\nMarks: " << marks;
    }
};
int main() {
    Student s;
    s.name = "Aarav"; s.rollNo = 12; s.marks = 85.5;
    s.display();
}
```

Output:

```
Name: Aarav
Roll No: 12
Marks: 85.5
```

Program 2: Rectangle area

```
#include <iostream>
using namespace std;
class Rectangle {
public:
    float length, breadth;
    float area() { return length * breadth; }
};
int main() {
    Rectangle r;
    r.length = 5; r.breadth = 3;
    cout << "Area: " << r.area();
}
```

Output:

```
Area: 15
```

Program 3: Product total price

```
#include <iostream>
using namespace std;
class Product {
public:
    int id, quantity; float price;
    float total() { return quantity * price; }
};
int main() {
    Product p;
    p.id = 101; p.quantity = 4; p.price = 25.5;
    cout << "Product ID: " << p.id
         << "\nTotal price: " << p.total();
}
```

Output:

```
Product ID: 101
Total price: 102
```

2. Object

Program 1: Create and use a Student object

```
#include <iostream>
using namespace std;
class Student {
public:
    string name; int roll;
    void show() { cout << name << " - Roll No: " << roll; }
};
int main() {
    Student s;
    s.name = "Mira"; s.roll = 7;
    s.show();
}
```

Output:

```
Mira - Roll No: 7
```

Program 2: Rectangle area and perimeter

```
#include <iostream>
using namespace std;
class Rectangle {
public:
    int l, b;
    int area() { return l * b; }
    int perimeter() { return 2 * (l + b); }
};
int main() {
    Rectangle r; r.l = 6; r.b = 4;
    cout << "Area: " << r.area()
         << "\nPerimeter: " << r.perimeter();
}
```

Output:

```
Area: 24
Perimeter: 20
```

Program 3: Even or odd using an object

```
#include <iostream>
using namespace std;
class Number {
public:
    int n;
    void check() {
        if (n % 2 == 0) cout << n << " is even";
        else cout << n << " is odd";
    }
};
int main() {
    Number x; x.n = 17; x.check();
}
```

Output:

```
17 is odd
```

3. cout <<

Program 1: Display student details

```
#include <iostream>
using namespace std;
int main() {
    cout << "Name: Riya\nAge: 18\nDepartment: CSE"
         << "\nCollege: ABC College";
}
```

Output:

```
Name: Riya
Age: 18
Department: CSE
College: ABC College
```

Program 2: Multiplication table

```
#include <iostream>
using namespace std;
int main() {
    int n = 5;
    for (int i = 1; i <= 10; i++)
        cout << n << " x " << i << " = " << n * i << "\n";
}
```

Output:

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Program 3: Arithmetic operations

```
#include <iostream>
using namespace std;
int main() {
    int a = 12, b = 4;
    cout << "Sum: " << a + b
         << "\nDifference: " << a - b
         << "\nProduct: " << a * b
         << "\nQuotient: " << a / b;
}
```

Output:

```
Sum: 16
Difference: 8
Product: 48
Quotient: 3
```

4. cin >>

Program 1: Input two numbers and add

```
#include <iostream>
using namespace std;
int main() {
    int a, b;
    cout << "Enter two integers: ";
    cin >> a >> b;
    cout << "Sum: " << a + b;
}
```

Output:

```
Enter two integers: 12 8
Sum: 20
```

Program 2: Input rectangle dimensions

```
#include <iostream>
using namespace std;
int main() {
    float l, b;
    cout << "Enter length and breadth: ";
    cin >> l >> b;
    cout << "Area: " << l * b;
}
```

Output:

```
Enter length and breadth: 5 3
Area: 15
```

Program 3: Positive, negative, or zero

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cout << "Enter a number: ";
    cin >> n;
    if (n > 0) cout << "Positive";
    else if (n < 0) cout << "Negative";
    else cout << "Zero";
}
```

Output:

```
Enter a number: -6
Negative
```

5. Data Members

Program 1: Employee annual salary

```
#include <iostream>
using namespace std;
class Employee {
public:
    int empId; string name; float salary;
    void showAnnualSalary() {
        cout << "Employee: " << name
             << "\nAnnual salary: " << salary * 12;
    }
};
int main() {
    Employee e; e.empId = 4; e.name = "Karan"; e.salary = 30000;
    e.showAnnualSalary();
}
```

Output:

```
Employee: Karan
Annual salary: 360000
```

Program 2: Circle area and circumference

```
#include <iostream>
using namespace std;
class Circle {
public:
    float radius;
    void calculate() {
        cout << "Area: " << 3.14 * radius * radius
             << "\nCircumference: " << 2 * 3.14 * radius;
    }
};
int main() {
    Circle c; c.radius = 2; c.calculate();
}
```

Output:

```
Area: 12.56
Circumference: 12.56
```

Program 3: Marks in three subjects

```
#include <iostream>
using namespace std;
class Marks {
public:
    int m1, m2, m3;
    void result() {
        int total = m1 + m2 + m3;
        cout << "Total: " << total
             << "\nAverage: " << total / 3.0;
    }
};
int main() {
    Marks m; m.m1 = 75; m.m2 = 80; m.m3 = 85; m.result();
}
```

Output:

```
Total: 240
Average: 80
```

6. Member Functions

Program 1: Calculator functions

```
#include <iostream>
using namespace std;
class Calculator {
public:
    int add(int a, int b) { return a + b; }
    int subtract(int a, int b) { return a - b; }
    int multiply(int a, int b) { return a * b; }
};
int main() {
    Calculator c;
    cout << "Add: " << c.add(8, 2)
         << "\nSubtract: " << c.subtract(8, 2)
         << "\nMultiply: " << c.multiply(8, 2);
}
```

Output:

```
Add: 10
Subtract: 6
Multiply: 16
```

Program 2: Check prime number

```
#include <iostream>
using namespace std;
class Number {
public:
    int n;
    void checkPrime() {
        bool prime = n >= 2;
        for (int i = 2; i * i <= n; i++)
            if (n % i == 0) prime = false;
        cout << (prime ? "Prime" : "Not prime");
    }
};
int main() {
    Number x; x.n = 13; x.checkPrime();
}
```

Output:

```
Prime
```

Program 3: Factorial using a member function

```
#include <iostream>
using namespace std;
class Number {
public:
    int n;
    long long factorial() {
        long long f = 1;
        for (int i = 1; i <= n; i++) f *= i;
        return f;
    }
};
int main() {
    Number x; x.n = 5;
    cout << "Factorial: " << x.factorial();
}
```

Output:

```
Factorial: 120
```

7. Arrays

Program 1: Display array elements

```
#include <iostream>
using namespace std;
class Array {
public:
    int a[5] = {4, 8, 2, 9, 1};
    void display() {
        for (int i = 0; i < 5; i++) cout << a[i] << " ";
    }
};
int main() { Array x; x.display(); }
```

Output:

```
4 8 2 9 1
```

Program 2: Find largest element

```
#include <iostream>
using namespace std;
class Array {
public:
    int a[5] = {14, 6, 22, 9, 17};
    int largest() {
        int max = a[0];
        for (int i = 1; i < 5; i++)
            if (a[i] > max) max = a[i];
        return max;
    }
};
int main() {
    Array x; cout << "Largest: " << x.largest();
}
```

Output:

```
Largest: 22
```

Program 3: Sum and average of array

```
#include <iostream>
using namespace std;
class Array {
public:
    int a[4] = {10, 20, 30, 40};
    void calculate() {
        int sum = 0;
        for (int i = 0; i < 4; i++) sum += a[i];
        cout << "Sum: " << sum << "\nAverage: " << sum / 4.0;
    }
};
int main() { Array x; x.calculate(); }
```

Output:

```
Sum: 100
Average: 25
```

8. Function Inside the Class

Program 1: Display student details

```
#include <iostream>
using namespace std;
class Student {
public:
    string name = "Neha"; int roll = 21;
    void display() {
        cout << "Name: " << name << "\nRoll: " << roll;
    }
};
int main() { Student s; s.display(); }
```

Output:

```
Name: Neha
Roll: 21
```

Program 2: Area inside Rectangle class

```
#include <iostream>
using namespace std;
class Rectangle {
public:
    int length = 8, breadth = 3;
    int area() { return length * breadth; }
};
int main() {
    Rectangle r; cout << "Area: " << r.area();
}
```

Output:

```
Area: 24
```

Program 3: Even or odd inside Number class

```
#include <iostream>
using namespace std;
class Number {
public:
    int n = 24;
    void check() {
        if (n % 2 == 0) cout << "Even";
        else cout << "Odd";
    }
};
int main() { Number x; x.check(); }
```

Output:

```
Even
```

9. Function Outside the Class

Program 1: Display using scope resolution

```
#include <iostream>
using namespace std;
class Student {
public:
    void display();
};
void Student::display() {
    cout << "Student: Aman\nRoll No: 15";
}
int main() { Student s; s.display(); }
```

Output:

```
Student: Aman
Roll No: 15
```

Program 2: Rectangle area outside class

```
#include <iostream>
using namespace std;
class Rectangle {
public:
    int length, breadth;
    int area();
};
int Rectangle::area() {
    return length * breadth;
}
int main() {
    Rectangle r; r.length = 7; r.breadth = 4;
    cout << "Area: " << r.area();
}
```

Output:

```
Area: 28
```

Program 3: Factorial outside the class

```
#include <iostream>
using namespace std;
class Number {
public:
    int n;
    long long factorial();
};
long long Number::factorial() {
    long long f = 1;
    for (int i = 1; i <= n; i++) f *= i;
    return f;
}
int main() {
    Number x; x.n = 4;
    cout << "Factorial: " << x.factorial();
}
```

Output:

```
Factorial: 24
```

10. Access Specifiers

Program 1: Private data with public functions

```
#include <iostream>
using namespace std;
class Student {
private:
    int marks;
public:
    void setMarks(int m) { marks = m; }
    void showMarks() { cout << "Marks: " << marks; }
};
int main() {
    Student s; s.setMarks(88); s.showMarks();
}
```

Output:

```
Marks: 88
```

Program 2: Public data member

```
#include <iostream>
using namespace std;
class Box {
public:
    int size;
};
int main() {
    Box b; b.size = 10;
    cout << "Size: " << b.size;
}
```

Output:

```
Size: 10
```

Program 3: Protected member through inheritance

```
#include <iostream>
using namespace std;
class Parent {
protected:
    int value = 25;
};
class Child : public Parent {
public:
    void show() { cout << "Protected value: " << value; }
};
int main() { Child c; c.show(); }
```

Output:

```
Protected value: 25
```

11. Default Constructor

Program 1: Default Student constructor

```
#include <iostream>
using namespace std;
class Student {
public:
    string name; int roll;
    Student() { name = "Rohan"; roll = 10; }
    void display() { cout << name << "\nRoll: " << roll; }
};
int main() { Student s; s.display(); }
```

Output:

```
Rohan
Roll: 10
```

Program 2: Default Rectangle constructor

```
#include <iostream>
using namespace std;
class Rectangle {
public:
    int length, breadth;
    Rectangle() { length = 5; breadth = 4; }
    int area() { return length * breadth; }
};
int main() {
    Rectangle r; cout << "Area: " << r.area();
}
```

Output:

```
Area: 20
```

Program 3: Default Number constructor

```
#include <iostream>
using namespace std;
class Number {
public:
    int n;
    Number() { n = 16; }
    void check() {
        cout << n << (n % 2 == 0 ? " is even" : " is odd");
    }
};
int main() { Number x; x.check(); }
```

Output:

```
16 is even
```

12. Parameterized Constructor

Program 1: Student parameterized constructor

```
#include <iostream>
using namespace std;
class Student {
public:
    int roll; string name; float marks;
    Student(int r, string n, float m) {
        roll = r; name = n; marks = m;
    }
    void display() {
        cout << name << "\nRoll: " << roll << "\nMarks: " << marks;
    }
};
int main() {
    Student s(3, "Isha", 91.5); s.display();
}
```

Output:

```
Isha
Roll: 3
Marks: 91.5
```

Program 2: Rectangle parameterized constructor

```
#include <iostream>
using namespace std;
class Rectangle {
public:
    int length, breadth;
    Rectangle(int l, int b) { length = l; breadth = b; }
    int area() { return length * breadth; }
};
int main() {
    Rectangle r(9, 2); cout << "Area: " << r.area();
}
```

Output:

```
Area: 18
```

Program 3: Circle parameterized constructor

```
#include <iostream>
using namespace std;
class Circle {
public:
    float radius;
    Circle(float r) { radius = r; }
    float area() { return 3.14 * radius * radius; }
};
int main() {
    Circle c(3); cout << "Area: " << c.area();
}
```

Output:

```
Area: 28.26
```

13. Single Inheritance

Program 1: Person and Student

```
#include <iostream>
using namespace std;
class Person {
public:
    string name; int age;
};
class Student : public Person {
public:
    int roll; float marks;
    void display() {
        cout << "Name: " << name << "\nAge: " << age
             << "\nRoll: " << roll << "\nMarks: " << marks;
    }
};
int main() {
    Student s; s.name = "Dev"; s.age = 18; s.roll = 11; s.marks = 82;
    s.display();
}
```

Output:

```
Name: Dev
Age: 18
Roll: 11
Marks: 82
```

Program 2: Animal and Dog

```
#include <iostream>
using namespace std;
class Animal {
public:
    void eat() { cout << "Animal eats\n"; }
};
class Dog : public Animal {
public:
    void bark() { cout << "Dog barks"; }
};
int main() {
    Dog d; d.eat(); d.bark();
}
```

Output:

```
Animal eats
Dog barks
```

Program 3: Shape and Rectangle

```
#include <iostream>
using namespace std;
class Shape {
public:
    int length, breadth;
};
class Rectangle : public Shape {
public:
    int area() { return length * breadth; }
};
int main() {
    Rectangle r; r.length = 6; r.breadth = 5;
    cout << "Area: " << r.area();
}
```

Output:

```
Area: 30
```